

Data Dictionary

Reserve Adequacy Threshold Analysis

1. Overview

Parameter	Value
Primary Source	Central Bank of Sri Lanka (CBSL)
Data Range	1995 - 2025 (varies by series)
Key Event	April 12, 2022 - Sovereign Default
Research Question	Reserve threshold signaling crisis inevitability

Crisis Timeline

Date	Event
2019-04-21	Easter Sunday bombings
2020-03-01	COVID-19 pandemic begins
2021-03-01	PBOC swap activated (\$1.5B)
2021-08-01	NFA turns negative
2022-04-12	Sovereign default announced
2023-03-20	IMF EFF approved

2. Benchmarks & Metrics

2.1 Import Cover Ratio

Import Cover (months) = Gross Reserves (USD) / Monthly Imports (USD)

Level	Threshold	Interpretation
Comfortable	≥ 6 months	Adequate buffer
IMF Minimum	≥ 3 months	Minimum acceptable
Warning	< 2 months	Elevated risk
Critical	< 1 month	Imminent crisis

2.2 Greenspan-Guidotti Ratio

GG Ratio = Gross Reserves (USD) / Short-term External Debt (USD)

Level	Threshold	Interpretation
Adequate	≥ 1.0	Can cover all ST debt
Near-breach	1.0 - 1.5	Elevated rollover risk
Breach	< 1.0	Cannot cover ST debt

2.3 IMF ARA Metric

ARA = 5% x Exports + 5% x M2(USD) + 30% x ST Debt + 15% x Portfolio Liab.

Component	Weight	Transformation
Annual Exports	5%	Sum quarterly x 4
Broad Money M2	5%	Convert LKR to USD
Short-term Debt	30%	Direct (quarterly)
Portfolio Liabilities	15%	Equity + Debt from IIP

Level	ARA Ratio
Comfortable	>= 150%
Adequate	100-150%
Breach	< 100%

2.4 Net Usable Reserves

Net Usable = Gross Reserves - PBOC Swap (\$1.5B) - Predetermined Drains

3. Column Definitions

3.1 Reserve Assets (reserve_assets_monthly_cbsl.csv)

Column	Definition	Unit
gross_reserves_usd_m	Total official reserve assets	USD mn
fx_reserves_usd_m	Foreign currency reserves	USD mn
imf_position_usd_m	Reserve position in IMF	USD mn
sdrs_usd_m	Special Drawing Rights	USD mn
gold_usd_m	Monetary gold holdings	USD mn
other_reserves_usd_m	Other reserve assets	USD mn

3.2 External Debt (external_debt_usd_quarterly.csv)

Column	Definition	Unit
govt_total_usd_m	Total government external debt	USD mn
govt_short_term_usd_m	Government ST external debt (GG denom.)	USD mn
govt_long_term_usd_m	Government LT external debt	USD mn
central_bank_usd_m	Central bank external debt	USD mn
total_short_term_usd_m	Combined ST external debt	USD mn

3.3 Trade & External Flows

Column	Definition	Unit
imports_usd_m	Monthly merchandise imports (CIF)	USD mn
exports_usd_m	Monthly merchandise exports (FOB)	USD mn
tourism_earnings_usd_m	Monthly tourism receipts	USD mn
remittances_usd_m	Workers remittance inflows	USD mn
cse_net_usd_m	Net CSE portfolio flows	USD mn

3.4 Monetary Aggregates

Column	Definition	Unit
reserve_money_m0_lkr_m	Reserve money (M0)	LKR mn
broad_money_m2_lkr_m	Broad money (M2) - for ARA calc	LKR mn
money_multiplier_m1	M1 money multiplier	Ratio

3.5 Net Foreign Assets

Column	Definition	Unit
nfa_monetary_auth_lkr_b	NFA of Monetary Authorities (CBSL)	LKR bn
nfa_comm_banks_total_lkr_b	NFA of Commercial Banks (Total)	LKR bn
nfa_comm_banks_dbu_lkr_b	NFA of Commercial Banks (DBUs)	LKR bn
nfa_comm_banks_obu_lkr_b	NFA of Commercial Banks (OBUs)	LKR bn

NFA = Foreign Assets - Foreign Liabilities. Negative NFA indicates liabilities exceed assets.

4. Data Coverage

Dataset	Freq	Start	End	File
Reserve Assets	Monthly	Nov 2013	Dec 2025	reserve_assets_monthly_cbsl.csv
Central Govt Debt	Quarterly	Dec 2000	Q3 2025	central_govt_debt_quarterly.csv
External Debt (USD)	Quarterly	Q4 2012	Q3 2025	external_debt_usd_quarterly.csv
IIP	Quarterly	Q4 2012	Q3 2025	iip_quarterly_2025.csv
Monetary Aggregates	Monthly	Dec 1995	Sep 2025	monetary_aggregates_monthly.csv
Monthly Imports	Monthly	Jan 2007	Nov 2025	monthly_imports_usd.csv
Monthly Exports	Monthly	Jan 2007	Nov 2025	monthly_exports_usd.csv
Tourism Earnings	Monthly	Jan 2009	Nov 2025	tourism_earnings_monthly.csv
Remittances	Monthly	Jan 2009	Nov 2025	remittances_monthly.csv
CSE Flows	Monthly	Jan 2012	Dec 2025	cse_flows_monthly.csv
NFA (Money Supply)	Monthly	Jan 2005	Dec 2017	money_supply_monthly_clean.csv
NFA + REER	Monthly	Jan 2021	Feb 2024	D14_reer_nfa.csv

Known Data Gaps

Gap	Mitigation
Gross Reserves pre-2013	Use IIP quarterly or CBSL Annual Reports
NFA breakdown pre-2021	Use aggregate NFA from money supply
Quarterly vs Monthly debt	Interpolate or quarterly-only analysis

5. CBSL Table References

Table	Description
Table 2.02	Monthly Exports
Table 2.04	Monthly Imports
Table 2.12	Outstanding External Debt
Table 2.14.1	Tourism Earnings
Table 2.14.2	Workers Remittances
Table 2.14.3	CSE Portfolio Flows
Table 2.15.2	Reserve Data Template (Historical)
Table 4.02	Monetary Aggregates
Table 4.11	Reserve Money, Multipliers, Velocity

Source: <https://www.cbsl.gov.lk/en/statistics/statistical-tables>